

2-to-4 Decoder with Active Low Enable

A 2:4 decoder takes two select inputs (A1, A0) and activates exactly one of four outputs (Y0–Y3) based on their binary value, but only while the Enable input is asserted. With an **active-low enable** (E'), the decoder is enabled when E' = 0 and disabled (all outputs forced inactive) when E' = 1. The outputs shown below are also **active low**, matching common parts such as the 74LS139.

Truth Table

E'	A1	A0	Y0	Y1	Y2	Y3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

When E' = 1 (disabled), all outputs are forced high (inactive) regardless of A1, A0 (marked X, don't care). When E' = 0 (enabled), exactly one output goes low — the one corresponding to the binary value of A1A0 — while the rest remain high.

Boolean Expressions

Each output is the NAND of an internal active-high enable signal E = NOT(E') with the appropriate literals of A1 and A0:

$$Y0 = \text{NOT}(E \cdot A1' \cdot A0')$$

$$Y1 = \text{NOT}(E \cdot A1' \cdot A0)$$

$$Y2 = \text{NOT}(E \cdot A1 \cdot A0')$$

$$Y3 = \text{NOT}(E \cdot A1 \cdot A0)$$

Note: if your outputs are active-high instead, flip the 0s and 1s in the Y columns above; the enable logic (E' = 0 to enable) stays the same.

Gate-Level Circuit Implementation

E' is inverted to produce the internal active-high enable E . $A1$ and $A0$ are each split into a direct line and an inverted line ($A1'$, $A0'$). Four 3-input NAND gates combine E with the correct pair of literals to produce each active-low output, per the equations above.

